7.Perform basic video processing operations on the captured video. Read captured video in

python and display the video, in slow motion and in fast motion.

Aim

To write a Python program to read a captured video and display it in normal speed, slow motion, and fast motion using OpenCV.

Procedure

1. Import the OpenCV library.
2. Open the input video file.
3. Check whether the video is opened successfully.
4. Read the video frame by frame.
5. Display the video in slow motion by increasing the delay between frames.
6. Display the video in fast motion by reducing the delay between frames.
7. Release the video and close all windows.

Code

```
import cv2

cap = cv2.VideoCapture("sample.mp4")

if not cap.isOpened():
   print("Video not found")
else:
   while True:
      ret, frame = cap.read()

      if not ret:
         break

      cv2.imshow("Slow Motion", frame)
      if cv2.waitKey(100) & 0xFF == ord('q'):
         break
```

```
    cap.release()

    cap = cv2.VideoCapture("sample.mp4")

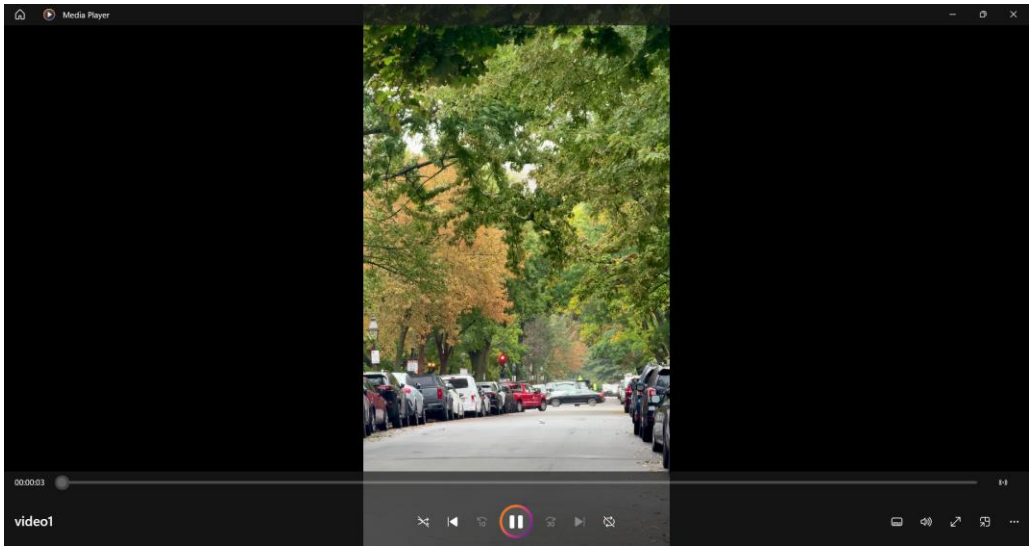
    while True:
        ret, frame = cap.read()

        if not ret:
            break

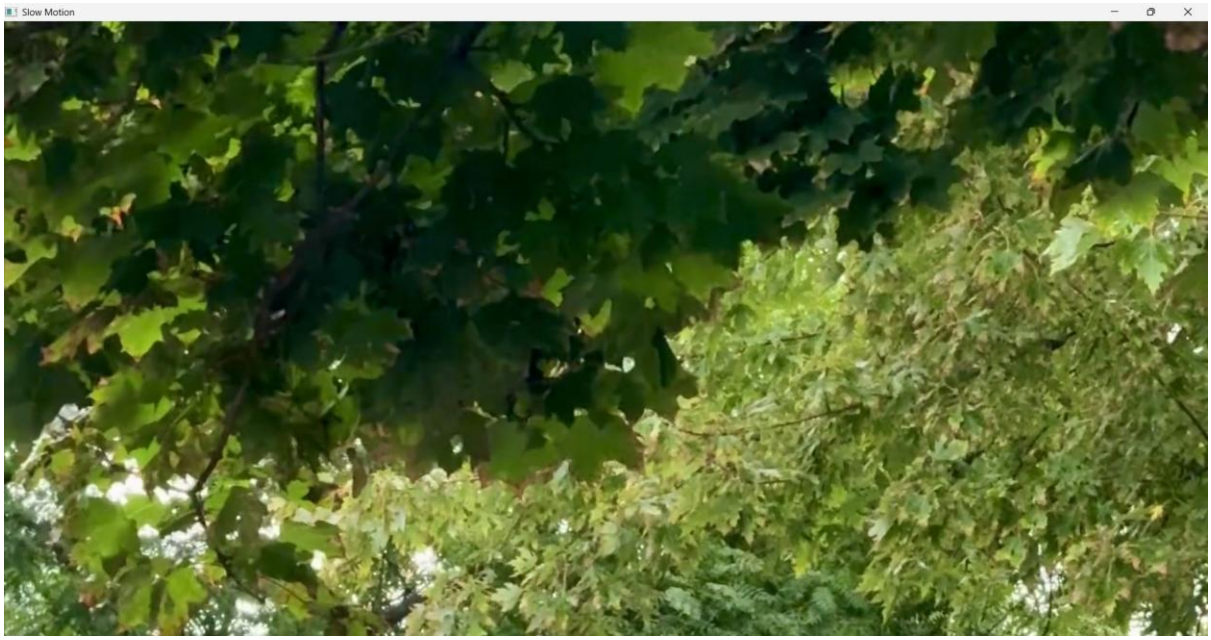
        cv2.imshow("Fast Motion", frame)
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break

    cap.release()
    cv2.destroyAllWindows()
```

SAMPLE INPUT:



SAMPLE OUTPUT:



Result

Thus, the captured video was successfully read and displayed in slow motion and fast motion using OpenCV. Press q to stop playback.